Machine Learning Algorithms in the Data Science Discourse Community

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Introduction

Data scientists nowadays incorporate machine learning into their workflows more than in the past due to its ability to handle massive datasets more quickly than traditional methods. As data becomes larger and more complex, older statistical approaches alone are often insufficient to process it in a timely manner. Machine learning models can handle large volumes of data rapidly and improve accuracy. At modern times, machine learning techniques have been applied across various industries, such as finance, agriculture, and logistics to enhance company systems. As new models and techniques in data science continue to emerge, complex problems are becoming easier and faster to solve.

Research Question:
"What role do machine learning algorithms play in the data science discourse community?"

The Data Science Discourse Community

Discourse communities are defined by Melzer (2020) as groups of individuals that share common goals and meet certain characteristics.

Discourse Community Characteristics:
The data science community fulfills six key features: have common public goals that are agreed among their members; utilize forms of communications; use these communications to share information and feedback; involve one or more genres to help reach the goal of their communities; acquire some specific lexis, jargon or terminology; and have a threshold level of expert members

Mechanisms of Communication:
Members communicate through GitHub to showcase work and projects, Discord channels for group conversations, online forums and blogs as Q&A platforms, and dashboards for visual representations of their work.

Specialized Language:
The community uses technical terminology including data wrangling, data cleaning, data mining, feature engineering, exploratory data analysis, regression, machine learning, neural networks, XGBoost, training data, validation data, cross validation, accuracy, confusion matrix, and evaluation tables.

Applications of Data Science

Finance & Business:
Companies improve efficiency and profitability through data science. Customer data analysis helps understand purchasing behavior and customize marketing strategies (Ullah et al., 2025).

Smart Cities:
ML models analyze data from street cameras, GPS, and traffic sensors to optimize traffic flow and reduce energy consumption (Ullah et al., 2025).

Big Data Processing:
Machine learning handles massive datasets more quickly than traditional methods, applied across finance, agriculture, and logistics.

Technology Advancement:
New models like Graph Neural Networks (GNNs) and Generative Adversarial Networks (GANs) solve complex problems faster (Luo et al., 2025).

References

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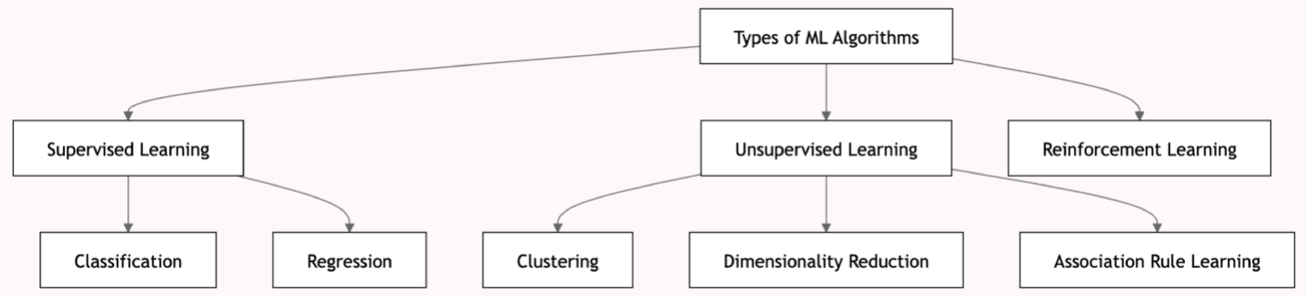
Machine Learning Algorithm Families

Supervised Learning
Models are trained using labeled data where each input has a corresponding output. Includes regression for numerical targets and classification for categorical outputs.
Examples: Linear regression, decision trees, random forest, SVM, KNN, neural networks, gradient boosting.

Unsupervised Learning
Works with unlabeled data to find patterns or relationships. Tasks include clustering and dimensionality reduction.
Examples: K-Means, hierarchical clustering, PCA, t-SNE.

Reinforcement Learning
An agent learns through rewards and penalties based on actions. It optimizes behavior to maximize long-term reward.
Example: AI in chess games.



Types of Machine Learning Algorithms

Evaluation Metrics

Classification

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN}$$
$$\text{Precision} = \frac{TP}{TP+FP}$$
$$\text{Recall} = \frac{TP}{TP+FN}$$
$$F_1 = \frac{2PR}{P+R}$$

Regression

$$\text{MAE} = \frac{1}{n}\sum |y_i - \hat{y}_i|$$
$$\text{MSE} = \frac{1}{n}\sum (y_i - \hat{y}_i)^2$$
$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$

Overview

Metric	Type
Accuracy	Class
Precision	Class
Recall	Class
F1 Score	Class
MAE	Reg
MSE	Reg
R^2	Reg

Classification Notes
- Accuracy: overall correctness
- Precision: exactness of positives
- Recall: coverage of positives
- F1: balance of precision/recall

Regression Notes
- MAE: average absolute error
- MSE: penalizes large errors
- R^2: variance explained

Interpretation
- Classification → discrete labels
- Regression → continuous values
- Choose metric based on task

Data Science Workflow

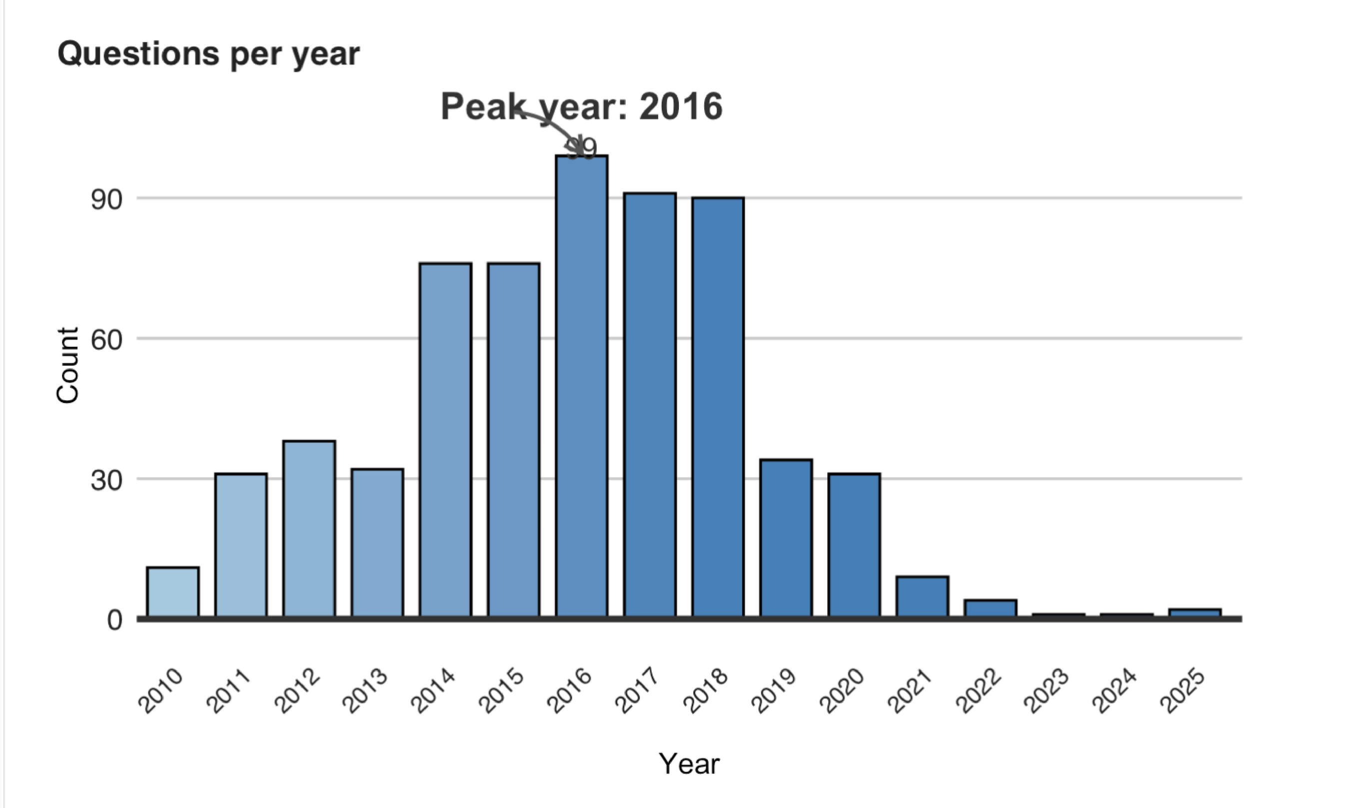
Objective Understanding → Data Mining → Data Cleaning → Data Exploration → Feature Engineering → Predictive Modeling → Data Visualization

Findings & Conclusion

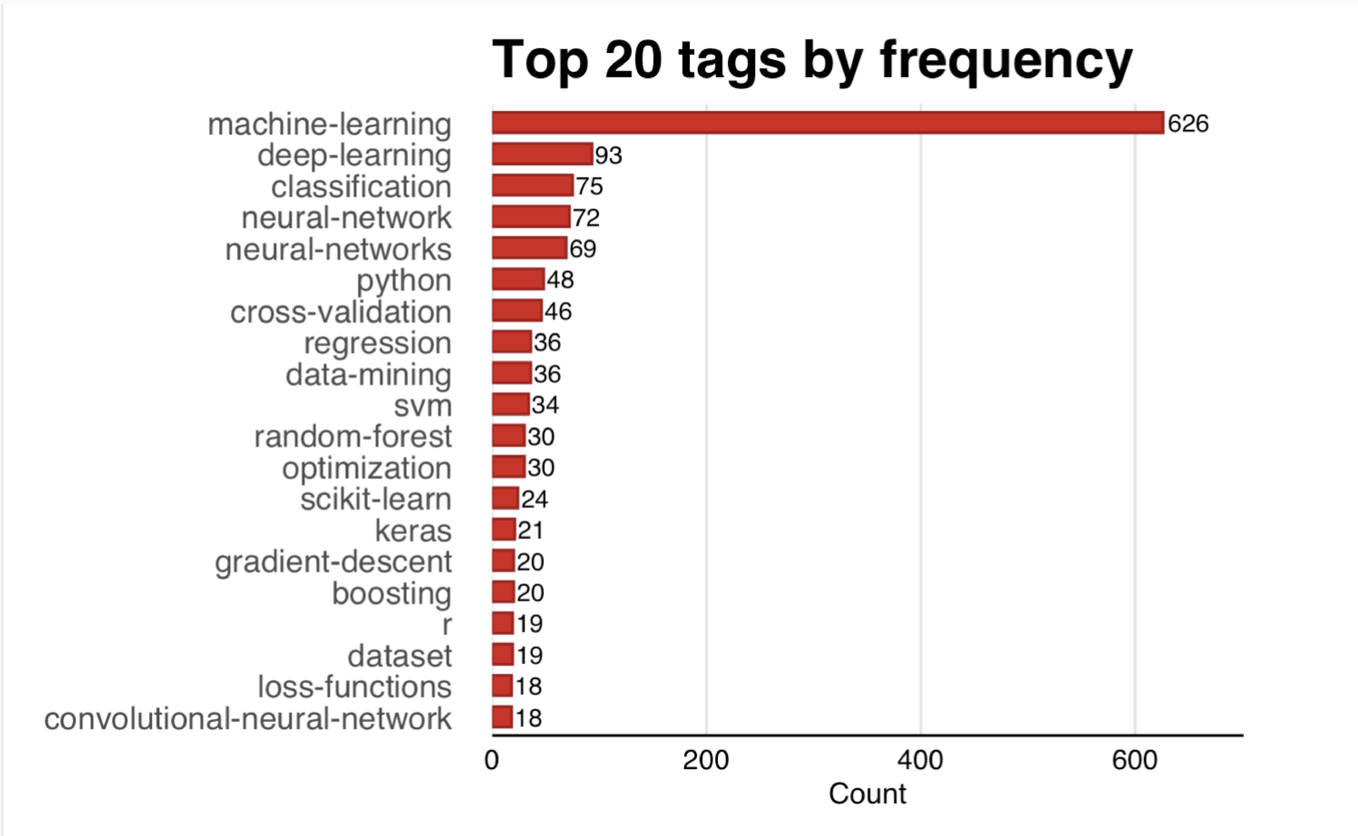
The texts differ in their focus across distinct stages of the data science workflow. These variations highlight that members in this discourse community tend to ask questions on specific steps rather than the whole workflow. The findings reveal that conceptual understanding still becomes the main question even until today. It is illustrated by how most of the questions that were asked by members in this discourse community concentrate on the "why" instead of just "what". Model evaluation and algorithms are the central questions in this community. Additionally, the discussions are mostly about machine learning both based on the highest scores and recent dates, which portrays how widely used machine learning is in this discourse community. The texts also depict that members put a heavy emphasis on interpretation and real world implementations on top of algorithms and model performance metrics, such as when handling issues like class imbalance. Overall, this community is centered towards both theories and applications. The bar charts reflect the primary focus of this community is on machine learning fundamentals, followed by algorithms and tools that are typically used by data scientists. The most non-active period of this discourse community appear to be from 2022-2025. This is mainly because during those years, artificial intelligence had been developed, which made some members prefer to ask their questions in artificial intelligence chat bots rather than in online forums.

Methodology & Primary Data

Data Source: Stack Exchange forum
Sample Size: 626 observational units
Analysis Tool: R Studio
Method: Text analysis
Methodology:
The research process involved four key steps: collecting raw data from Stack Exchange, cleaning data by handling missing values, exploring through graphical and numerical summaries, and interpreting visualizations for general audience.



Top 10 Tags by Frequency



Top 20 Tags Distribution

Top Machine Learning Questions

By Score	By Date (2022-2025)
What is the difference between test set and validation set?	Use of training data that has been labeled by the AI model itself
How to understand the drawbacks of K-means	In training a neural network, why don't we take the derivative with respect to the step size in gradient descent?
"Why is Euclidean distance not a good metric in high dimensions?"	Why does data science see class imbalance as a problem for supervised learning when statistics does not?
ROC vs precision-and-recall curves	How does an LLM parameter relate to a weight in a neural network?
How to know that your machine learning problem is hopeless?	What is the difference between something being true and true with probability 1?